

Deep learning theory: Lecture 5

Recap: Consider the Gradient Descent update

$$\underline{\theta_{k+1} = \theta_k - \gamma_k D_{\theta} R(\theta_k)} \quad (GD)$$

and the corresponding Gradient Flow approximation

$$\begin{cases} \frac{d}{dt} \bar{\theta}_t = -D_{\theta} R(\bar{\theta}_t) \\ \bar{\theta}_0 = \theta_0 \end{cases} \quad (GF)$$

Then if $\gamma_k = \gamma \ll 1$ and $D_{\theta} R$ is L -Lipschitz cont.

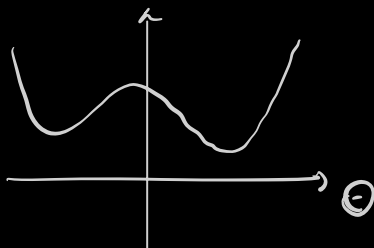
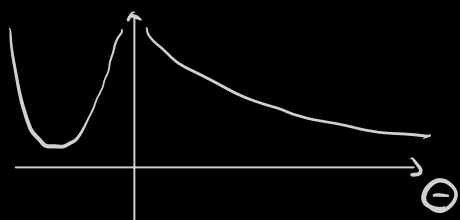
Lemma: Let $D_{\theta} R$ be Lipschitz. For every $T > 0$ there exists $C > 0$ s.t

$$\|\bar{\theta}_{k\gamma} - \theta_k\| \leq C\gamma \quad \forall k \in \{0, \dots, \lfloor \frac{T}{\gamma} \rfloor\}$$

Q: when does (GF) converge, and when can we bound

$$\hat{R}(f_{\theta}) - \inf_{f \in \mathcal{F}} \hat{R}(f) \quad ?$$

One can easily prove that $R(f_{\theta_t})$ (not θ_t) has a limit as $t \rightarrow \infty$



However, we can't say much about optimality of limit points.

Example (linear regression) $n \geq d$ $X^T X \succeq \lambda \mathbb{1}$

$$\hat{R}(\theta) = \frac{1}{n} \|X\theta - y\|_2^2 \implies \theta_* = (X^T X)^{-1} X^T y$$

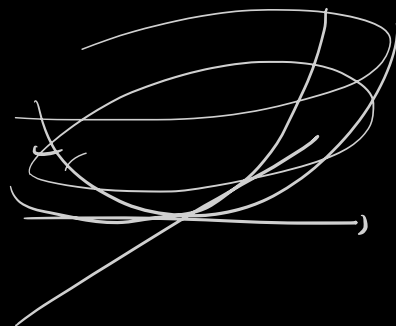
$$\implies R(\theta_*) = \frac{1}{n} \|X(X^T X)^{-1} X^T \mathbb{1} y\|_2^2$$

$$\begin{aligned} \frac{d}{dt} \|\theta_t - \theta_*\|_2^2 &= \langle \theta_t - \theta_*, -D_\theta \hat{R}(\theta_t) \rangle = -\langle \theta_t - \theta_*, X^T (X\theta_t - y) \rangle \\ &= -\langle \theta_t - \theta_*, X^T X (\theta_t - \theta_*) \rangle \leq -\lambda \|\theta_t - \theta_*\|_2^2 \end{aligned}$$

$$\stackrel{G}{\implies} \|\theta_t - \theta_*\|_2^2 \leq e^{-\lambda t} \|\theta_0 - \theta_*\|_2^2 \quad \text{☺}$$

This happens because $\hat{R}$ is strongly convex:

$$\begin{aligned} \hat{R}(\theta) - \hat{R}(\theta_*) &= D_\theta R(\theta_*) \cdot (\theta - \theta_*) + \frac{1}{2} (\theta - \theta_*)^T D_\theta^2 R(\theta_*) (\theta - \theta_*) \\ &= \frac{1}{2} (\theta - \theta_*)^T X^T X (\theta - \theta_*) \end{aligned}$$



Thm (Grönwall inequality) Let $u: [0, \infty) \rightarrow \mathbb{R}$ be differentiable and

$$\frac{d}{dt} u_t \leq \gamma u_t$$

$$\text{then: } u_t \leq u_0 e^{-\gamma t}$$

The convergence properties of GF are guaranteed when R is convex:

Def: a differentiable $\hat{R}: \Theta \rightarrow \mathbb{R}$ is λ -strongly convex for $\lambda \in \mathbb{R}$ if:

$$(*) \quad \hat{R}(\theta') \geq \hat{R}(\theta) + \langle D_{\theta} \hat{R}(\theta), \theta' - \theta \rangle + \frac{\lambda}{2} \|\theta' - \theta\|^2 \quad \text{for any } \theta, \theta' \in \Theta$$

Lemma: If R is convex (λ -strongly convex with $\lambda=0$) and $DR(\theta_*)=0$, $\theta_* \in \min R$

Pf: trivial from definition.

furthermore if $\lambda > 0$
the minimum is unig.

Lemma let $\hat{R}$ be λ -convex, then the following holds:

$$\langle D_{\theta} \hat{R}(\theta') - D_{\theta} \hat{R}(\theta), \theta' - \theta \rangle \geq \lambda \|\theta - \theta'\|^2$$

Proof: Consider:

$$\hat{R}(\theta') \geq \hat{R}(\theta) + \langle D_{\theta} \hat{R}(\theta), \theta' - \theta \rangle + \frac{\lambda}{2} \|\theta - \theta'\|^2$$

adding the above to the definition of λ strongly convex we are done $\square$

Thm: Let R be diffable, λ -strongly convex and θ_* be the minimizer, then

$$\|D_{\theta} R(\theta)\|_2^2 \geq 2\lambda (R(\theta) - R(\theta_*)) \quad (\text{Łojasiewicz})$$

Proof: Setting $\theta' = \theta - \frac{1}{\lambda} D_{\theta} R(\theta)$ in $(*)$ we get

$$R(\theta_*) \geq R(\theta) - \frac{1}{\lambda} \|D_{\theta} R(\theta)\|^2 + \frac{1}{2\lambda} \|D_{\theta} R(\theta)\|^2 = R(\theta) - \frac{1}{2\lambda} \|D_{\theta} R(\theta)\|^2$$

Thm: Let $R(\theta)$ be λ -convex, ^{L-smooth.} then there exists a unique θ^* and

$$\|\bar{\theta}_t - \theta_*\|^2 \leq \|\bar{\theta}_0 - \theta_*\|^2 e^{-2\lambda t}$$

$$R(\bar{\theta}_t) - R(\theta_*) \leq (R(\bar{\theta}_0) - R(\theta_*)) e^{-2\lambda t}$$

Proof: $\frac{d}{dt} \frac{1}{2} \|\bar{\theta}_t - \theta_*\|^2 = \langle \bar{\theta}_t - \theta_*, \frac{d}{dt} \bar{\theta}_t \rangle$

$$= -\langle \bar{\theta}_t - \theta_*, D_{\theta} R(\bar{\theta}_t) - D_{\theta} R(\theta_*) \rangle$$

$$\leq -\lambda \|\bar{\theta}_t - \theta_*\|^2$$

→ Gronwall

Similarly $\frac{d}{dt} R(\bar{\theta}_t) - R(\theta_*) = \langle D_{\theta} R(\bar{\theta}_t), \dot{\bar{\theta}}_t \rangle$

$$= -\|D_{\theta} R(\bar{\theta}_t)\|^2$$

Łojasiewicz →

$$\leq -2\lambda (R(\bar{\theta}_t) - R(\theta_*))$$

→ Gronwall

The convergence of GD can also be established.

Lemma: Assume that R is L -smooth, λ strongly convex.

Choose $\eta = \frac{1}{L}$. Then, there exists $\kappa \in \mathbb{R}$ s.t.

$$R(\theta_t) - R(\theta_*) \leq (R(\theta_0) - R(\theta_*)) e^{-\frac{t}{\kappa}}$$

Proof: $R(\theta_t) = R(\theta_{t-1} - \frac{1}{L} D_{\theta} R(\theta_{t-1}))$

L -smoothness → $\leq R(\theta_{t-1}) - \frac{1}{L} \|D_{\theta} R(\theta_{t-1})\|^2 + \frac{1}{2L} \|D_{\theta} R(\theta_{t-1})\|^2$

$$\Rightarrow R(\theta_t) - R(\theta_*) \leq R(\theta_{t-1}) - R(\theta_*) - \frac{1}{2L} \|D_{\theta} R(\theta_{t-1})\|^2$$

Łoj. → $\leq (1 - \frac{1}{2L}) (R(\theta_{t-1}) - R(\theta_*))$

□

Combining the two results above we have uniform in time convergence

Lemma: Let $\bar{\theta}_t$ converge to θ_* and there exists $N(\theta_*)$ s.t

$$D^2 R(\theta) \succeq \delta^2 I \quad \forall \theta \in N$$

$$\lim_{\delta \rightarrow 0} \sup_t \|\bar{\theta}_t - \theta_t^{(n)}\| = 0$$

Pf: Let T be such that $\bar{\theta}_t \in N$.

We have:

$$\sup_{t \in \infty} \|\bar{\theta}_t - \theta_t^{(n)}\| = \max \left(\sup_{t \leq T} \|\bar{\theta}_t - \theta_t^{(n)}\|, \sup_{t > T} \|\bar{\theta}_t - \theta_t^{(n)}\| \right)$$

$$\text{we know that for every fixed } T_n, \lim_{\delta \rightarrow 0} \sup_{t \leq T} \|\bar{\theta}_t - \theta_t^{(n)}\| = 0$$

then it is sufficient to show that for every $\varepsilon > 0$ there exists finite T_n :

$$\lim_{\delta \rightarrow 0} \sup_{t \leq T_n} \|\bar{\theta}_t - \theta_t^{(n)}\| < \varepsilon$$

To show this, we show that there exists a sufficiently small ball $B_{\varepsilon'}(\theta_*)$ such that when the dynamics enter $B_{\varepsilon'}(\theta_*)$ they never leave $B_{\frac{\varepsilon}{2}}(\theta_*)$

Since $\frac{d}{dt} R(\bar{\theta}_t) < 0$ and by above lemma for θ_* it is sufficient to show that:

$$\inf_{y \in \partial B_{\frac{\varepsilon}{2}}(\theta_*)} R(y) > \sup_{y \in B_{\varepsilon'}(\theta_*)} R(y)$$

which holds since

$$\inf_{y \in \partial B_{\frac{\varepsilon}{2}}(\theta_*)} R(y) - R(\theta_*) \geq \delta^2 \inf_{y \in \partial B_{\frac{\varepsilon}{2}}(\theta_*)} \|y - \theta_*\|^2 = \delta^2 \left(\frac{\varepsilon}{2}\right)^2 = \delta^2 \left(\frac{\varepsilon}{2\varepsilon'}\right)^2 \sup_{y \in B_{\varepsilon'}(\theta_*)} \|y - \theta_*\|^2$$

$$\geq \frac{\delta^2}{L} \left(\frac{\varepsilon}{2\varepsilon'}\right)^2 \sup_{y \in B_{\varepsilon'}(\theta_*)} R(y) - R(\theta_*)$$

□

Gradient flows for convex functionals. recall: $X^T X \vartheta^* = X^T y \quad \forall \vartheta^*$

Example: Consider now the case where some eigenvalues of $X^T X$ are 0.

$$\frac{d}{dt} \vartheta_t - \vartheta_* = -X^T X (\vartheta_t - \vartheta_*) \implies \vartheta_t - \vartheta_* = e^{-X^T X t} (\vartheta_0 - \vartheta_*)$$

$$R(\vartheta_t) - R(\vartheta_*) = \left(\frac{1}{2} (\vartheta_t - \vartheta_*)^T X^T X (\vartheta_t - \vartheta_*) \right) = \frac{1}{2} (\vartheta_0 - \vartheta_*)^T \underbrace{e^{-X^T X t} X^T X e^{-X^T X t}}_{A(t)} (\vartheta_0 - \vartheta_*)$$

$$\text{Eig}(A(t)) = e^{-\lambda t} \cdot \lambda \cdot e^{-\lambda t} = \frac{1}{2t} (2\lambda t) e^{-2\lambda t} \leq \frac{1}{t}$$

Theorem: Let $R(\vartheta)$ be convex and Lipschitz smooth. then



$$R(\bar{\vartheta}_t) - R(\vartheta_*) \leq C \frac{1}{t}$$

Proof: $R(\vartheta') \geq R(\vartheta) - D_\vartheta R(\vartheta)(\vartheta - \vartheta')$

$$\begin{aligned} \implies \frac{d}{dt} \frac{1}{2} \|\vartheta_t - \vartheta_*\|^2 &= (\vartheta_t - \vartheta_*)^T (-D_\vartheta R(\vartheta_t)) \\ &\leq R(\vartheta_t) - R(\vartheta_*) \end{aligned}$$

$$\implies \int_0^t R(\bar{\vartheta}_s) - R(\vartheta_*) ds \leq -\frac{1}{2} \|\vartheta_t - \vartheta_*\|^2 + \frac{1}{2} \|\vartheta_0 - \vartheta_*\|^2 \leq \frac{1}{2} \|\vartheta_0 - \vartheta_*\|^2$$

$$\implies \frac{1}{t} \int_0^t R(\bar{\vartheta}_s) ds - R(\vartheta_*) \leq \frac{1}{2t} C_0$$

$$\text{but } \frac{1}{t} \int_0^t R(\bar{\vartheta}_s) ds \geq R(\vartheta_t) \text{ since } \frac{d}{dt} R(\vartheta_t) \leq 0$$

$$\implies R(\bar{\vartheta}_t) - R(\vartheta_*) \leq \frac{1}{2t} C_0$$

□

Stochastic Gradient Descent

In practice, the evaluation of the gradient

$$\begin{aligned} D_{\theta} \hat{R}(\theta_+) &= D_{\theta} \left(\frac{1}{n} \sum_{j=1}^n e(f_{\theta_+}(x_j), y_j) \right) \\ &= \frac{1}{n} \sum_{j=1}^n D_{\theta} e(f_{\theta_+}(x_j), y_j) \cdot D_{\theta} f_{\theta_+}(x_j) \end{aligned}$$

is quite expensive for large n .

Often, people evaluate this gradient on a minibatch

$B_+ \subseteq \{1, \dots, n\}$ of size $b_+ \ll n$ (a sample of $\{1, \dots, n\}$ of size b_+)

$$\widetilde{D_{\theta} R}(\theta_+) = \frac{1}{b_+} \sum_{j \in B_+} D_{\theta} e(f_{\theta_+}(x_j), y_j) \quad (\text{SG})$$

Note that this is an unbiased estimator of $D_{\theta} \hat{R}(\theta_+)$:

$$\mathbb{E}(\widetilde{D_{\theta} R}(\theta_+)) = \mathbb{E}\left(\frac{1}{b_+} \sum_{B_+} D_{\theta} e(f_{\theta_+}(x_j), y_j)\right)$$

sample w. replacement $\searrow$ $= \frac{1}{b_+} b_+ \mathbb{E}(D_{\theta} e(f_{\theta_+}(x_j), y_j))$

$$= \frac{1}{n} \sum_{j=1}^n D_{\theta} e(f_{\theta_+}(x_j), y_j).$$

with covariance matrix (for $b_+ = 1$)

$$\frac{1}{n} \sum_{j=1}^n (\bar{D}_\theta e(f_{\theta_t}(x_j), y_j) - D\hat{R}(\theta_t)) \otimes (\bar{D}_\theta e(f_{\theta_t}(x_j), y_j) - D\hat{R}(\theta_t))$$

Then, the Stochastic Gradient descent update reads:

$$\begin{aligned} \theta_{t+1} &= \theta_t + \gamma_t \tilde{D}_\theta \tilde{R}_t(\theta_t) \quad (\text{SGD}) \\ &= \theta_t + \gamma_t D_\theta \hat{R}(\theta_t) + \gamma_t (\tilde{D}_\theta \tilde{R}_t(\theta_t) - D_\theta \hat{R}(\theta_t)) \end{aligned}$$

Analogously to what was done in the GD setting, we want to approximate the above with the solution of a differential equation.

In this case, given the stochastic nature of the update, the natural candidate for a limiting equation is a stochastic differential equation of the form.

$$d\bar{\theta}_t = b(\bar{\theta}_t) dt + \sigma(\bar{\theta}_t) dB_t$$

where $b(\bar{\theta}_t) \in \mathbb{R}^d$ is the drift coefficient

$\sigma(\bar{\theta}_t) \in \mathbb{R}^{d \times p}$ is the diffusion coefficient

B_t is a p -dimensional Brownian motion.